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Lab #3
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4201 L

Lab 3: Measurements, Error and Statistics and EXTRA CREDIT

Introduction

This lab investigates how the method used to conduct measurement affects data of resistance values. 100 resistors of 15 k Ω with 5% resistance were used, and three different measurement systems were used: a bench Tek 4020 (2-wire), a handheld EM830 DMM, and the Tek 4020 in 4-wire (Kelvin) mode. After gathering the data, we examine how instrument resolution and the system used influences the mean, spread (standard deviation), and histogram shape of the measurements. Predictions of the system configuration influences on the data can be seen in step 1, and actual results and confirmation of the predictions can be seen in step 4. In step 1, I formulated a prediction about the variation and effects of using each system, which was found to be correct in step 4. I correctly predicted that the Tek 4020 showed a narrower, smoother normal distribution with smaller deviation, higher accuracy due to higher resolution, and a mean closer to 15kohms. Using the em830 led to a lower accuracy and wider spread (due to worse resolution) and very slightly varied from the normal distribution in the Tek 4020 data. There were no outliers outside of the acceptable range of the 5% tolerance, and no indication of systematic error (besides minor 2 wire system added resistance) and no outliers. More information about the statistical elements of resistance is covered in step 4. Additionally, the impacts and logistics of using a 2 wire system versus a 4 wire system will be explored, for measurement of lower resistance values. Finally, various specialized meter equipment will be discussed, identified and used for extra credit.

1. Measure and record in a spreadsheet the values for a set of around 100 resistors given to you by the instructor using two measurement systems, 15k 1% or 5%. ETE4201 F25 values: Team up with a partner(s) to make the measurements, only one R type (1% or 5%) per person/group is needed. After being cut from the resistor roll, mark them such that the measurements can be repeated or checked easily, and the instructor will explain a method. When not measuring them carefully, fold them and put them in an envelope taking care so they do not detach from the tape roll so they can be reused. Before making measurements answer this question, writing brief comments below: 1. What do you expect to find in terms of the range of resistance values and

any variation in resistor spread, i.e. amount of resistance spread, numbers of resistors in or out of spec. Etc.. For the following measurements make sure you can match corresponding resistor measurements from each test, meaning R1 is the same physical resistor in each case by labeling the resistors.

To answer the question prompt, I want to analyze the difference between measuring a set of resistors with 1% vs 5% tolerance and then analyze the difference between measuring with the two different systems. I predict to see a bell curve for 1% or 5% with either system. . The mean for both 1% and 5%, there would be a mean of 15k, median of 15k, and a normal distribution. As far as the 1% resistors, we would expect to see an approximate range of $[15k\Omega - (15k\Omega * 1\%)]$ to $[15k\Omega + (15k\Omega * 1\%)]$ or a range of 14.85 ohms to 15.15 ohms on our bell curve. For the 5% resistors, in a perfect world we would see a range of $[15k\Omega - (15k\Omega * 5\%)]$ to $[15k\Omega + (15k\Omega * 5\%)]$, or a range of 14.25 ohms to 15.75 ohms. Anything outside of that range is considered an outlier. I think that the 15k ohm 100 resistors for 1% would have a much narrower spread around 15k, and a bit of a wider spread in the 5%. Also I predict that the actual error for 5% will be less than 5%, because of industry standards. I would predict that this will be the case similarly for 1%, but the actual error would probably be closer to 1% as compared to the 5%. Generally I think resistors will fall into this range and that there will be no outliers despite whatever measurement system might be used, maybe not even in the entire class. The range and spread of data will not only be influenced by choosing to use 5% instead of 1%, but also be influenced by using a higher quality meter versus a lower quality DMM with less decimal points. The Tek 4020 will show a narrower, smoother normal distribution with smaller deviation, higher accuracy due to higher resolution, and a mean closer to 15kohms. The second system, using the em830, will have a lower accuracy and wider spread due to worse resolution, less digits shown, worse standard for calibration and accuracy, probably vary *very slightly* from an exact normal distribution as compared to the other system, and possibly lead to rounding errors skewing the data. However I think there still won't be any outliers outside of the range I defined and be within tolerance.

2 & 3. System 1: Measure resistors using the Tek 4020 Multimeter in the lab using 2-wire resistance measurement techniques.

System 2: Using any personal Multimeter, typically a 3.5-digit meter (this can be done in lab or outside but keep your resistors the same so you can check one measurement system to another. Record meter type/model. Also record the approximate time it takes to make the measurements.

Meter type/ model: DMM em830

Approximate measurement time: 8.33 minutes

System 3: Measure a few select resistors using the Tek 4020 Multimeter in the lab using 4-wire resistance measurement techniques, check a few measurements and see if these are consistent with part 1

	Theoretical	Tek 4020	DMM em830						
1 15k		14.994	15.07		theretical	2 wire	4 wire	kelvin rs	kelvin rp
2 15k		14.999	15.29	1 1 ohm		1.067	0.986	0.9913	0.991
3 15k		15.008	15.07	2 2 ohm		1.068	0.991	0.9932	0.9932
4 15k		14.784	15.11	3 3 ohm		1.062	0.987	0.9897	0.9897
5 15k		14.959	15.06	4 4 ohm		1.059	0.985	0.9926	0.9927
6 15k		15.029	15	5 5 ohm		1.059	0.989	0.9894	0.9894
7 15k		15.054	15.19						
8 15k		15.064	15.04					1khz	
9 15k		15.11	15.28					rs=resistar series	
10 15k		14.981	15.05					rp=resistar parallel	
11 15k		15.082	15.06						
12 15k		15.201	15.1	Variables from photos					
13 15k		15.256	15.04	rs					
14 15k		15.002	15.04	rp					
15 15k		15.046	15.06	cs					
16 15k		15.056	15.05	cp					
17 15k		15.1	15.05	lp					
18 15k		14.951	15.04	ls					
19 15k		15.076	15.01	Q					
20 15k		15.017	15.04						
21 15k		15.267	15.4	Measurements of DMM values took 8.33 minutes					
22 15k		15.031	15.05						
23 15k		15.101	15.02						
24 15k		15.108	15.03						
25 15k		15.119	15.03						
26 15k		14.952	15.26						
27 15k		15.129	15.02						
28 15k		15.06	15.05						

IMG 1. Spreadsheet of the resistance values for all three systems.

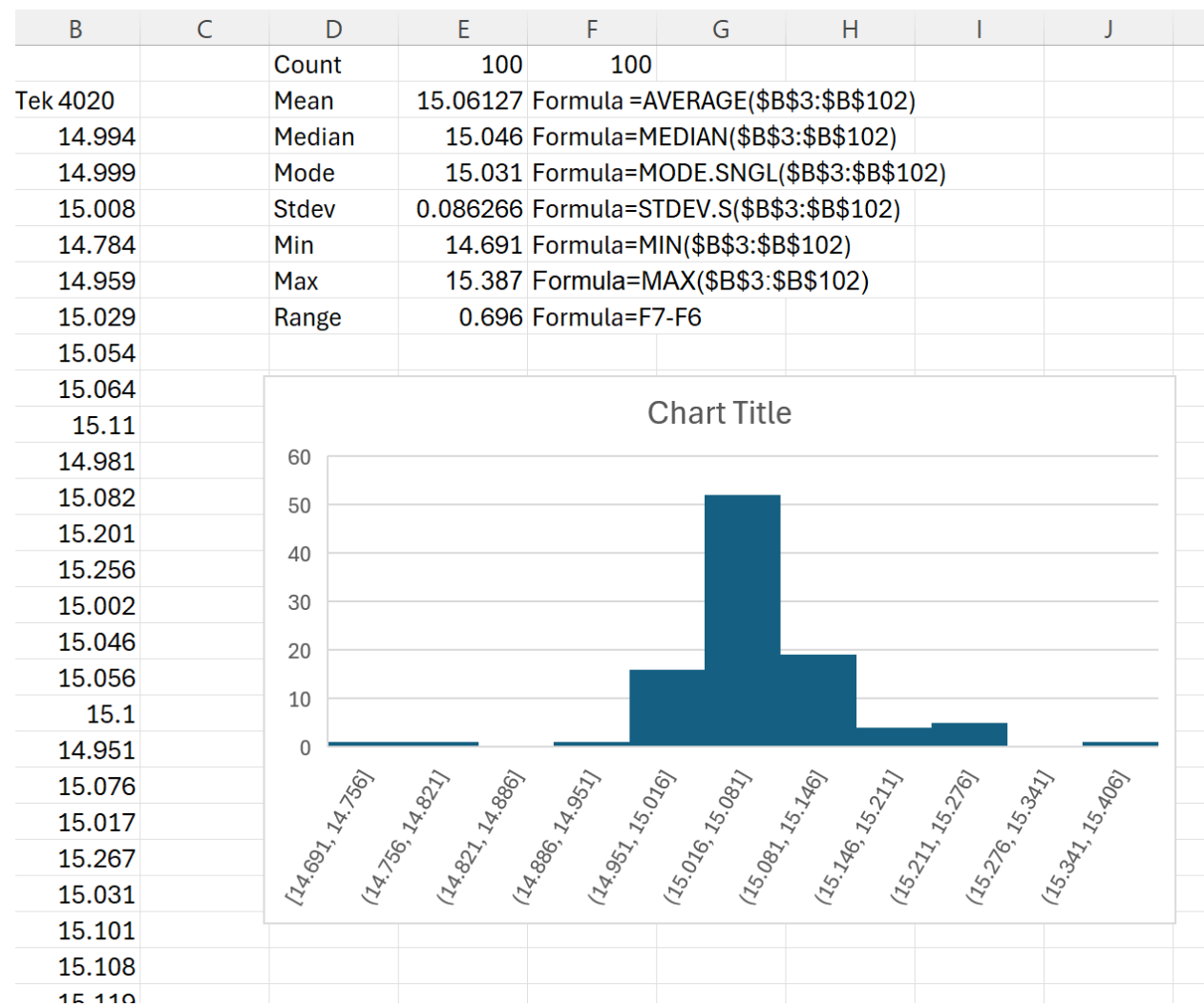
All data from this question can be found in the spreadsheet, accessible using this link.

https://onedrive.live.com/:x:/g/personal/D55F45C705EE916D/ETArjAcgJj5JndDU3ukTDFsBkQtSsYlr4r1aHJKOxjtEyA?resid=D55F45C705EE916D!s078c2b302620493e9dd0d4dee9130c5b&ithint=file%2Cxlsx&e=4ZNppo&migratedtospo=true&redeem=aHR0cHM6Ly8xZHJ2Lm1zL3gvYy9kNTVmNDVjNzA1ZWU5MTZkL0VUQXJqQWNnSmo1Sm5kRFUzdWtUREZzQmtRdFNzWUlyNHlxYUhKS094anRFeUE_ZT00Wk5wcG8

3. Use MSEXcel spreadsheet to compute significant statistics (mean, median, mode and standard deviation) for the resistors and create a histogram of the data. Use the MSEXcel help files or Lynda.com statistics tutorial to guide you on how to do this.

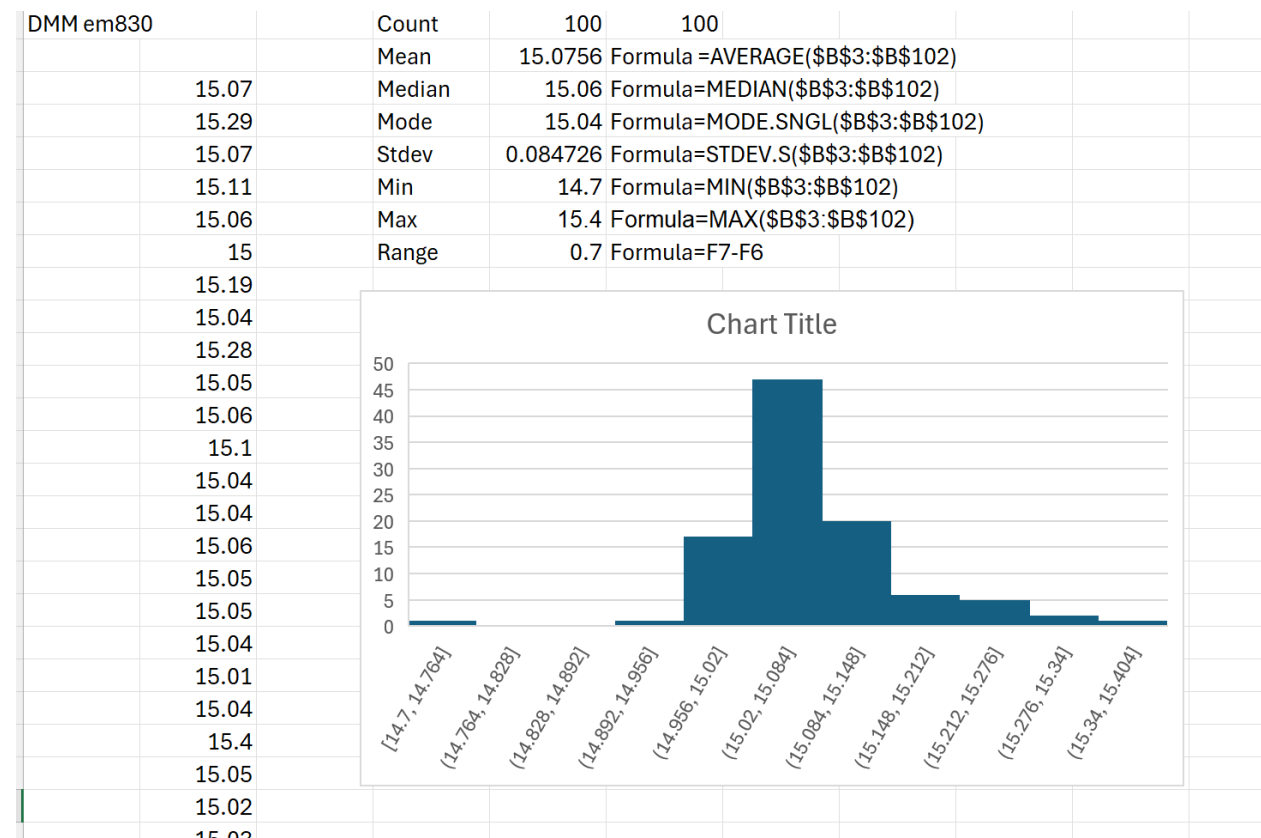
References: <https://en.wikipedia.org/wiki/Statistics> /Normal distribution /Median etc

System 1:



IMG 2.

System 2:



IMG 3.

4. Compare the measured values from each system i.e. R1 (measurement 1) to R1 (measurement 2) to R1 (measurement 3) to see if the values differ consistently or not, this would likely indicate some systematic error or any outlier data. Did your measurements have any outliers or other data issues?

Surprisingly, system 1 and 2 varied very minimally. For system 2, the mean was 15.8k as compared to the mean of 15.6k for system 1, indicating that the tek4020 data is slightly closer to the theoretical value of 15k. All data was in the predicted range for 5% tolerance resistance values: $[15k\Omega - (15k\Omega * 5\%)]$ to $[15k\Omega + (15k\Omega * 5\%)]$, or a range of 14.25 ohms to 15.75 ohms. The other statistical results like range, mode, min, max and deviation are incredibly similar, and even just increasing the sample size could eliminate these minor differences. There were no outliers outside of the acceptable range of the 5% tolerance, and no indication of systematic error (besides minor 2 wire system added resistance) and no outliers.

5. Briefly comment on your results and how they might have differed from your original premise. Explore and comment on any other statistics principles (i.e. within MSExcel)

My original premise, found on page 2, was correct. I already elaborated on additional statistical principles.

6 & 7. This was from last week but many did not seem to recognize how the 4 wire meter worked so if your data provides you the source value for a given resistor range you need not have to do this part. Use a second multimeter to determine the I source current used by the meter to measure resistance. Do this by setting your meter under test to ohms then insert an ammeter into the circuit with the resistance, does the resistance value on the main meter change with the 2nd meter in the circuit, take measurements for 100 ohms, 1000 ohms 10,000ohms, 100k and near 1M with the decade resistance box. Use a 4wire setup/meter to measure a number of 1ohm resistors and compare the 4Wire and 2Wire resistance measurement values. Examine the Kelvin clip method of making a 4Wire meter measurement.

System 3:

	theretical	2 wire	4 wire
1 1 ohm		1.067	0.986
2 2 ohm		1.068	0.991
3 3 ohm		1.062	0.987
4 4 ohm		1.059	0.985
5 5 ohm		1.059	0.989

IMG 4.

Analysis:

A DMM measures ohms by forcing a known current through a section of the circuit and measuring the resulting voltage, then using ohms law in the form $R = \frac{V}{I}$. This is one reason why it's critical to disconnect the source voltage when measuring the resistance. In two wire systems, the same two leads carry the test current and sense the voltage. This means that the reading of the resistance also includes extra resistance from the test leads and contact resistance. An extra

few ohms when measuring large amounts like mega ohms is not a huge deal in most cases, but when measuring small amounts of resistance (like 5 ohms) a few extra ohms can skew data in major ways. In 4-wire (Kelvin) systems, two leads carry current and two separate leads measure the voltage at the resistor body. This means that the extra added resistance from the leads and contact is eliminated. Essentially we are separating the current source from the voltage measurement and thus eliminating resistance from leads. This is a great system to use when measuring small amounts of resistance, when extra long leads are needed or when there's a need for extreme precision. But most of the time, this extra precaution is unneeded. In this case, we are measuring a resistance of less than 10 ohms, so the difference from using the Kelvin system is quite visible. The measurements from the two wire system are *somewhat* consistent with the 4 wire system, but the 4 wire system data points are preferable and more precise.

8. A number of specialized meters will be in the lab including nanovoltmeter, picometers, Electrometers and Sourcemeters. Record their equipment numbers and find their manuals, specifications on the web.

Specialized test gear includes component testers with 4-wire (Kelvin) clips—two leads for current, two for voltage—to eliminate lead resistance and get accurate low-ohm readings. We can actually use them to also measure capacitance and inductance! Ultra-sensitive instruments like picoammeters, nanovoltmeters, and electrometers can read especially tiny currents and voltages. A source-measure unit (SMU) is like a 4-wire system with a programmable supply that can source and measure voltage or current over wide ranges, and not just the small test current a DMM provides. It can supply any current and voltage up to a given limit, and has much more power than a 4 wire meter because a 4 wire meter only can put out a certain current for a measurement while an SMU can put out almost any current. Also, many SMUs have 4 wire clips and can do some basic statistics. It can do bins for histograms (such as taking in data and putting it in 1%, 2% in) and can build data analysis for mass quantities in real time thus saving time and increasing efficiency in some situations.



IMG 5. pico



IMG 6. SMU



IMG 7. Electrometers



IMG 8. nano

9. Extra Credit use a SourceMeter to setup a measurement or Phillips Automatic RCL to measure your resistors, this unit has built in statistics for “binning” of helping sort components, this is not often done for common circuits but for specialized instruments and low count units (<50) this can be a useful practice

We used a Gw Instek lcr -6002 precision lcr meter and adjusted the configuration of resistance in series and resistance in parallel, and could see the change in capacitance and inductance. We recorded the change in the variables r_s , r_p , c_s , c_p , l_p , l_s , Q .

	theretical	2 wire	4 wire	kelvin r_s	kelvin r_p
1	1 ohm	1.067	0.986	0.9913	0.991
2	2 ohm	1.068	0.991	0.9932	0.9932
3	3 ohm	1.062	0.987	0.9897	0.9897
4	4 ohm	1.059	0.985	0.9926	0.9927
5	5 ohm	1.059	0.989	0.9894	0.9894

IMG 9.



Conclusion

In this lab I measured 100 resistors of 15k ohms in three different systems. I analyzed how data changes statistically based on tolerance (1% versus 5%) and the three different systems used in the lab (Tek 4020 Multimeter, DMM em830, Tek 4020 with 4-wire configuration). In step 1, I formulated a prediction about the variation and effects of using each system, which was found to be correct in step 4. I correctly predicted that the Tek 4020 showed a narrower, smoother normal distribution with smaller deviation, higher accuracy due to higher resolution, and a mean closer to 15kohms. Using the em830 lead to a lower accuracy and wider spread (due to worse resolution) and very slightly varied from the normal distribution in the Tek 4020 data. There were no outliers outside of the acceptable range of the 5% tolerance, and no indication of systematic error (besides minor 2 wire system added resistance) and no outliers. More information about the statistical elements of resistance are covered in step 4. Additionally, the impacts and logistics of using a 2 wire system versus a 4 wire system was explored- and it was determined that using a 4 wire system is ideal for measuring resistance values less than 10 ohms. Finally, various specialized meter equipment was discussed and identified, and practice using the specialized equipment was done for extra credit.

